Andy Zhu

US Citizen

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EDUCATION

Rutgers University - New Brunswick, NJ *May 2027*

Bachelor of Science in Computer Science, Minor in Mathematics

- **Honors and Certifications**: Dean's List 2023/24, Java Oracle Certified Foundations Associate 2022, CompTIA Fundamentals+ 2023, Intuit QuickBooks Online Accounting Certification 2023

SKILLS & COURSEWORK

Languages: TypeScript, JavaScript, Python, SQL, Java, C++, PHP
Frameworks and Tools: React, Next.js, TailwindCSS, ShadCN, Supabase, PostgreSQL, Git, GitHub, Linux, Vercel
Coursework: Data Structures and Algorithms, Discrete Math, Computer Architecture, Software Methodology, Database Management Systems, Linear Optimization, Physics I,II, Calculus I-IV, Introduction to Data Science, Design of Algorithms

PROFESSIONAL EXPERIENCE

Aersys Inc. **Piscataway, NJ**

Engineering Intern *August 2026 – Present*

- Engineering a low-cost, proprietary camera equipment platform that replicates the functionality of systems typically priced near $2,000 at a target cost of roughly $200, cutting hardware costs by approximately 90%.
- Developing custom firmware for ESP32 microcontrollers to drive core device functionality, integrating embedded software with physical hardware components.
- Building and assembling functional prototypes end-to-end, including 3D-printed enclosures, circuit wiring, and soldering, bridging embedded systems development with hands-on hardware fabrication.

Austin ECHO **Remote**

Lead Software Engineering Intern *May 2026 – August 2026*

- Led a full website redesign within a cross-functional team, coordinating design workflows in Figma and Notion with development and deployment on WordPress.
- Implemented front-end and back-end functionality using JavaScript, PHP, and MySQL, improving site structure, content management, and overall performance.
- Integrated Google Analytics tracking to monitor SEO visibility and CTR for future user data based decisions.

Rutgers IEEE - VEXU **New Brunswick, NJ**

President & Lead Software Engineer *September 2023 – Present*

- Led a 10-member engineering team through Agile-style development cycles, managing sprint planning, code reviews, and a multi-branch GitHub repository to maintain code stability.
- Directed development of a scalable robotics codebase, implementing core autonomous subsystems including differential drive control, wheel odometry, color sorting, and macro systems.
- Tuned custom PID controllers and motion profiling, improving autonomous path accuracy by 35% and execution speed by 60% during competition trials.

Rutgers IEEE (ieee.rutgers.edu) **New Brunswick, NJ**

Webmaster *April 2025 – Present*

- Reconstructed the Rutgers IEEE website using a modern React based architecture, significantly improving long term maintainability, developer control, and reducing average page load time by ~35%.
- Built and deployed fully responsive UI components, improving mobile usability and layout consistency.

PROJECTS

RU Open -ruopen.vercel.app **New Brunswick, NJ**

Software Engineering - Full Stack Web Development *August 2025 - August-2025*

- Developed a fullstack classroom availability platform using Next.js, TypeScript, Node.js, PostgreSQL, and Supabase, allowing students to check real time room occupancy.
- Designed and implemented relational database schemas supporting fast lookups and scalable future expansion.
- Built responsive UI components using ShadCN and React Server Components, improving usability across both mobile and desktop devices.

Portfolio Website - azhu.dev **New Brunswick, NJ**

Software Engineering - Front End Web Development *July 2025 – August 2025*

- Built a personal developer portfolio using React and TypeScript, featuring a modern bento grid layout with interactive cards and smooth animations.
- Implemented dark mode, custom components, and responsive design for consistent UX across desktop and mobile.
- Showcased projects, technical skills, and interactive UI elements to improve recruiter visibility and portfolio engagement.